You can preinstall and preconfigure plug-ins (data sources and visualizations) on Grafana startup. The details are beyond the scope of the book, but as a starting point, here's an example. Let's say you want to preinstall the ClickHouse data source plug-in (assuming you have the `grafana-clickhouse-ds` binary available locally) with ClickHouse running self-hosted or from one of the SaaS offerings. Using Docker Compose, you'd add the following lines to the docker-compose.yaml YAML manifest in the Grafana section:

```
...
    volumes:
    - ./ds.yaml:/etc/grafana/provisioning/datasources/datasource.yaml
    environment:
      GF_INSTALL_PLUGINS: grafana-clickhouse-ds
```

The binary of the plug-in, in your local filesystem

In addition, you would define the ClickHouse data source in a file called ds.yaml (which is used in the Docker Compose file previous) with the following content:

```
apiVersion: 1
datasources:
  - name: ClickHouse
    type: grafana-clickhouse-datasource
    jsonData:
      defaultDatabase: default
      port: 9000
      server: clickhouse
      username:
      tlsSkipVerify: true
      secureJsonData:
        password:
```

Defines the type of data source plug-in

Below, the data source plug-in-specific configuration

If you now do a `docker-compose up`, the ClickHouse data source is automatically added and configured, readily available for you to use in Grafana.

After this short detour on how to provision plug-ins, let's have a look at a concrete data source. After clicking the Add Data Source button (I picked CloudWatch; see figure 6.1, noting that I trimmed out certain lines, so you will see more here), you should see data source–specific configuration options, including authentication, authorization, and various defaults or limits you want to set.

In this view, you can also define whether the data source should be used as the default data source for new dashboards or alerts (the Default toggle is in the upper-right corner), and if you click the Dashboards tab, you will usually (but not always) find predefined dashboards for the data source. You can use those predefined dashboards as is or as a starting point for your own dashboards. Finally, in the bottom row, you will find several buttons, and with the Save & Test button, Grafana performs a simple (data source–dependent) check to see if it can access the data source with the credentials you provided.

NOTE The test that Grafana carries out when you press the Save & Test button is typically very basic. Think of it as a necessary but not sufficient